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Operation optimization of existing industrial circulating water system considering variable frequency drive

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ABSTRACT

The traditional circulating cooling water system regulates the flow rate through parallel pumps and valves. Due to fluctuations in operation, there will be lots of energy waste, resulting in inefficient operation of the circulating cooling water system. This research proposed a mixed-integer nonlinear programming (MINLP) method. By installing variable frequency drives (VFDs) on the pumps, changing constant-speed pumps (CSPs) into variable speed pumps (VSPs), we can regulate flow rate with valves and VSPs simultaneously. Optimizing the system using Genetical algorithm (GA) to obtain precise operating parameters, including the configuration of VFDs, the number of CSPs and VSPs and the resistance coefficients of valve in the branch and main pipe. Sensitivity analysis of the model is carried out, and the energy-saving effect of VFD under different temperature differences, pump types and electricity prices, the operating status of the pump and the status of the pipeline are compared. The results indicate that the MINLP genetic algorithm optimization model for industrial circulating water considering variable frequency drive proposed in this paper has a wide range of applicability. It can save an average of 23 % of the total cost under different conditions, and has a promising energy saving effect.

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1. Introduction

In chemical production, circulating cooling water systems (CCWSs) are designed for the rejection of waste heat to the environment. The energy consumption of CCWSs accounts for a large proportion of CCWSs (Zheng et al., 2018). Of the world's total energy generated, pumping systems, especially the centrifugal pumps consume about 20 % (Arun Shankar et al., 2016). Therefore, the research is of great significance to the optimization of the CCWSs. Cooling towers, circulating water pumps and pipelines are all crucial components of the CCWSs. Quite a few articles mainly study the optimization problem of each component in the system when designing

the CCWSs, including cooling towers (Gololo and Majozi, 2011), pump system (Liu et al., 2019), cooling networks (Varbanov et al., 2018) and the interactions between them. The main function of the cooling tower is to make the circulating water whose temperature rises to exchange heat with the external environment. Usually, there will be a large amount of water loss. The function of the pump system is to transport the circulating water to each operation unit, such as transporting the circulating water to the heat exchange unit to exchange heat with the hot stream, and complete the cooling task. To optimize the design of the CCWSs, the researchers propose a series of mathematical models. Kim and Smith, (2001) first proposed a mathematical model that emphasizes interaction between cooling tower and cooler network. Ponce-Ortega et al. (2007) proposed a MINLP model that considers the capital costs of both coolers and cooling water. In addition, in order to relieve the heat load pressure of the

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Nomenclature

A	The heat exchange area of the heat exchanger, m ² .
A _j	Parameter of the performance curve of pump j, dimensionless.
B _j	Parameter of the performance curve of pump j, dimensionless.
C _j	Parameter of the efficiency curve of pump j, dimensionless.
c _p	Heat capacity, J/(kg·°C).
d	Inner diameter, m.
D _j	Parameter of the efficiency curve of pump j, dimensionless.
e	Roughness of the pipe, m.
E _j	Parameter of the efficiency curve of pump j, dimensionless.
F _j	Parameter of the efficiency curve of pump j, dimensionless.
g	Acceleration of gravity, kg·m/s ² .
H	Supply head of pump, m.
k	Speed ratio of variable speed pump, dimensionless.
K	Overall heat transfer coefficient, W/(m ² ·K).
l	Length, m.
m	Number of heat exchange tubes, dimensionless.
M	Mass flowrate, kg/s.
n	Sub-period month n, dimensionless.
N	Number of tube passes, dimensionless.
NTU	Number of Transfer Units, dimensionless.
ΔP	Pressure drop, Pa.
Q	Heat load of heat exchanger, W.
R	Ratio of the smaller fluid heat capacity flow rate to the larger heat capacity flow rate of the heat exchanger, dimensionless.
T	Temperature, °C.
u	Velocity, m/s.
V	Flowrate, m ³ /s.

Greek symbols

α	Convection transfer coefficient, W/(m ² ·K).
ε	Heat transfer efficiency of heat exchanger, dimensionless.
ζ	Local resistance coefficient of heat exchanger, dimensionless.
η	Efficiency of pump, dimensionless.
λ	Friction coefficient, dimensionless.
ξ	Local resistance coefficient, dimensionless.
ρ	Density of cooling water, kg/m ³ .
ω	0–1variable, the opening of pump, dimensionless.

Subscript

branch	The branch of pipe.
ex	Exchanger.
i	Heat exchanger i.
j	Pump j.
t	Cooling tower.

The above article mainly describes a series of designing optimization problems of CCWSs. However, the operation of the CCWSs will deviate from the design, and many design parameters, such as ambient temperature, air humidity and electricity prices, will change. Therefore, in addition to optimizing the system when designing the CCWSs, optimizing the operating process can additionally reduce energy waste. [Castro et al. \(2000\)](#) reduce the operating cost of the CCWSs by minimizing the flow rate. [Wang et al. \(2017\)](#) considered the adjustment of velocity through the split ratio optimization to change the velocity in each heat exchanger. [Souifi and Souissi, \(2019\)](#) proposed a process integration approach based on Pinch analysis technique to address energy efficiency issues in cooling water networks for retrofit design problem. [Arsenyeva et al. \(2013\)](#) studied the scaling problem of circulating water in the heat exchanger.

However, these articles only reformed the CCWSs, and did not consider installing a VFD on the pump to optimize the system. In recent years, in order to reduce the energy consumption of the pump system, more articles have begun to focus on the energy saving effect of installing VFDs, but the purchase and installation costs of VFDs is still high. Recent works focus on minimizing power consumption of pumps and of cooling tower fans ([Pontes et al., 2022](#)). The synergy between high cooling water flow rates and slack values increases the difficulty of the optimal design. Often, flow rates are significantly above 5,000 m³·h⁻¹, and to comply with these values, large pumps, motors and VFDs would be required ([Pontes et al., 2022](#)). Adjusting the rotational speed of centrifugal pumps through VFDs can save much energy ([Arun Shankar et al., 2016](#)). There are currently few studies on optimizing the number of pumps and the number of frequency converters in a water pumping system, ([Koor et al., 2014](#)) optimally determine the number of opening and closing of a series of identical VSPs within the best efficiency point (BEP) range of the pump. [Xia et al. \(2015\)](#) present a novel method to optimize the number of variable speed pumps and presented the economic effect of variable speed pumps number. [Harish et al. \(2010\)](#) developed a theoretical model that established the viability of using VSPs and CSPs in power plants, which will help CCWSs designers decide whether to use VSPs. [Pontes et al. \(2022\)](#) built a mathematical model to calculate the total energy of the pump system through total head loss, cooling water demand, pump model selected, and VFD use. This only considers the optimization of the water pump system, but does not consider the interconnection between the pipeline system and the variable frequency water pump system. The opening of the valve in the cooling network pipeline will affect the performance of the pipeline, as well as the intersection point of pump performance curve and the pipeline characteristic curve.

To deal with the above-mentioned research gap, this study proposes a complex mixed-integer nonlinear programming (MINLP) model based on Genetic Algorithm (GA) to optimize the existing industrial CCWSs, and the purpose is to minimize the cost of the pumping system (including capital and operating costs), taking into account heat transfer requirements and piping characteristics. Determine the opening and closing of a series of parallel pumps in the pump system, the number of VSPs and CSPs, and the speed ratio k of VSPs within the BEP range of the pump as well as the adjustment of valve opening of the main pipeline and branch. Sensitivity analysis is carried out under different

system, air coolers are installed in the CCWSs to share the heat load of the heat exchanger [Carvalho et al., \(2019\)](#), [Manassaldi et al. \(2014\)](#) presents a disjunctive mathematical model for the optimal design of air cooled heat exchangers.

temperatures differences, pump types and combinations, and electricity prices under valve control and VFD control.

2. Method

2.1. Problem statement

In a cooling water system, the cooling water is transported by the pump system to different pipeline branches, and it enters the heat exchanger to complete the heat exchange task. After the temperature rises, the cooling water enters the cooling tower and exchange heat with the environment to lower the temperature. For this system, the main energy consumption comes from the electrical energy consumption of the pump, and the electrical energy consumption of the pump mainly comes from the pressure drop of the pipeline, the height of the cooling tower, the pressure drop of the heat exchanger and its branch, and these pressure drops mainly depend on the flow of the cooling water, the valve opening and other factors. The increase in the flow rate of circulating water means an increase in the pressure drop in the pipeline system, and a decrease in the opening of the valve will also increase the resistance coefficient of valve in the pipeline, which will also increase the pressure drop, resulting in energy loss.

In this operational problem of the CCWSs, the heat load of the heat stream, the inlet and outlet temperature and the inlet temperature of the circulating water are known, and the ambient temperature of different months is different, that is, the temperature of the circulating water is different. In order to meet the heat demand of the heat exchanger, the model should configure the pump system and the pipeline system, that is, determine whether the pump is installed with a VFD, and whether the pump is running, valve opening of the pipeline, and determine the operating point through the performance curve of the pumps and the pipeline to provide circulating water for the heat exchanger. The objective function is to minimize the operating costs of the pump and the capital cost of installing the VFDs. Based on the optimization of genetic algorithm, a complex MINLP model is established in this paper to minimize the total cost and obtain the precise operating parameters of the CCWSs, the configuration of VFDs, the number of CSPs and VSPs and the resistance coefficients of valve in the branch and main pipe.

The flowchart of this work is shown in Fig. 1.

2.2. Model formulation

The goal of this work is to control the flow rate with the VFDs control and valve control to minimize the cost of the CCWSs. In the objective function, P_e (\$/kWh) is the price of electricity per kilowatt-hour. $V(n,j)$ (m^3/s) is the flow rate provided by the pump j in sub-period n , $\Delta p(n)$ (kPa) is the total pressure drop of the pipeline in sub-period n , and η_j is the efficiency of the pump j . n_{VFD} is the numbers of VFDs, and P_{VFD} (\$) is the capital cost of one VFD, and i is an annualized interest rate, and n_{year} is the year in which the appliance is simulated to run, and n_{pump} is the number of pumps.

$$obj = P_e \cdot \Delta P(n) \sum_{j=1}^{n_{pump}} \frac{V(n,j)}{\eta_j} + n_{VFD} \cdot P_{VFD} \cdot \frac{i(i+1)^{n_{year}}}{(1+i)^{n_{year}} - 1} \quad (1)$$

The price of installing a VFD on a pump is related to the rated power of the pump, so the capital cost of installing the VFD can be expressed by Eq.(2), where P_{rated} (kW) is the rated

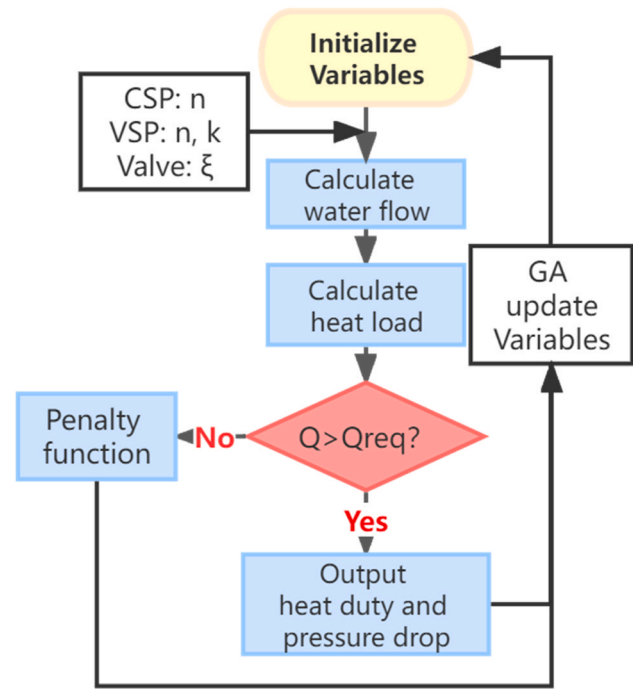


Fig. 1 – Flowchart of the proposed framework.

power of the pump, and P_{unit} (\$/kW) is the cost of VFD per unit rated power pump.

$$P_{VFD} = P_{rated} \cdot P_{unit} \quad (2)$$

In the objective equation, P_e is a constant, $V(n)$ and $\Delta P(n,i)$ is obtained by combining the characteristic curves of the pump and the pipeline, and the total pressure drop of the pipeline can be expressed by Eq.(3). In Eq.(3), $\Delta p_p(n,i)$ (kPa) is the pressure drop of the main pipeline in sub-period n , and $\Delta p_{ex}(n,i)$ (kPa) is the pressure drop of the heat exchanger i in sub-period n , and $\Delta p_{branch}(n,i)$ (kPa) is the pressure drop controlled by the valve in the branch of the heat exchanger i in sub-period n , and $H_t(m)$ is the height of the cooling tower, and $H_e(m)$ is the maximum height of the heat exchangers. The characteristic curve of the pipe can be represented by Eq. (4), and in Eq.(4) ρ (kg/m^3) is the density of circulating water, and g ($kg \cdot m \cdot s^{-2}$) is gravitational acceleration, and both are constants.

$$\Delta P(n,i) = \Delta p_p(n,i) + \Delta p_{ex}(n,i) + \Delta p_{branch}(n,i) + \rho g (H_t + H_e) \quad (3)$$

$$H(n,i) = \frac{\Delta P(n,i)}{\rho g} \quad (4)$$

In Eq.(3), total pressure drop $\Delta P(n,i)$ consists of four parts, the pressure drop of main pipeline $\Delta p_p(n,i)$, the pressure drop of heat exchanger $\Delta p_{ex}(n,i)$, pressure drop of the branch where the heat exchanger is located $\Delta p_{branch}(n,i)$, and the pressure drop caused by the height of the cooling tower and heat exchanger. The first three pressure drops are all functions of the flow rate.

For a pump, its characteristic curve can be represented by Eq.(5), where A and B are constants.

$$H(n) = A_j + B_j V^2 \left(n, i \right) \quad (5)$$

Its efficiency η_j in Eq.(1) can be expressed by Eq.(6), where C, D, E, F are constants in relation to the properties of the pump.

$$\eta_j = C_j + D_j \cdot V \left(n, j \right) + E_j \cdot V^2 \left(n, j \right) + F_j \cdot V^3 \left(n, j \right) \quad (6)$$

According to the law of proportion, the speed of the pump is changed, and the characteristic curve becomes an Eq.(7).

$$H(n) = A_j k^2 \left(n, j \right) + B_j V^2 \left(n, i \right) \quad (7)$$

The characteristic curve of the paralleling pumps can be expressed as Eq.(8), where $w(n, j)$ is the variable 0–1, which is used to indicate the opening and closing of the pump j , if $w(n, j) = 0$, it means that the pump j is closed in sub-period n , if $w(n, j) = 1$, it means pump j is working. The total flow rate can also be written as the sum of the flow rates provided by all pumps, or the sum of the flow rates of all the branches where the heat exchangers are located, expressed in Eq. (9) and Eq. (10), respectively, where i is the heat exchanger i and j is the pump j .

$$V(n) = \sum_{j=1}^n \omega \left(n, j \right) \sqrt{\frac{H(n) - A_j k^2 \left(n, j \right)}{B_j}} \quad (8)$$

$$V(n) = \sum_{i=1}^{n_{\text{exchanger}}} V(n, i) \quad (9)$$

$$V(n) = \sum_{j=1}^{n_{\text{pump}}} V(n, j) \quad (10)$$

Since all pumps are connected in parallel, the head of the pumps are same and can be expressed by Eq.(11).

$$H(n, i) = H(n) \quad (11)$$

2.2.1. Supply pressure model

In Eq.(3), $\Delta P_p(n)$ is the pressure drop of the main line, and $\Delta P_{ex}(n)$ is the pressure drop of the heat exchanger, and $\Delta P_{branch}(n)$ is the pressure drop of the branch where the heat exchanger is located, and λ is the coefficient of friction and calculated by Eq.(18), and $l(m)$ is the length of the main pipe, and $d(m)$ is the inner diameter of the main pipe.

$$\Delta P_p(n) = \left(\lambda \frac{l}{d} + \xi_{\text{main}}(n) \right) \frac{\rho u^2(n)}{2} \quad (12)$$

$$\Delta P_{ex} \left(n, i \right) = \lambda \frac{l_{ex}}{d_{ex}} \frac{\rho u^2(n, i)}{2} \cdot N(i) + \xi \frac{\rho u^2(n, i)}{2} \cdot N(i) + 1.5 \frac{\rho u^2(n, i)}{2} \quad (13)$$

$$\Delta P_{branch}(n) = \xi \left(n, i \right) \frac{\rho u_{branch}^2(n, i)}{2} \quad (14)$$

the flow rate in the heat exchanger pipe can be written into an equation according to the flow balance (15).

$$\frac{\pi d_{ex}^2}{4} u \left(n, i \right) \frac{m}{N} = V \left(n, i \right) \quad (15)$$

The flow rate in the main pipe can be written as an Eq.(16).

$$u(n) = \frac{4V(n)}{\pi d^2} \quad (16)$$

The flow rate of the circulating water in the branch where the heat exchanger is located $u_{branch}(m/s)$ can be expressed as Eq.(17), where $d_{branch}(m)$ is the inner diameter of the pipe where the heat exchanger is located.

$$u_{branch} \left(n, i \right) = \frac{4V(n, i)}{\pi d_{branch}^2} \quad (17)$$

The coefficient of friction can be calculated by Eq.(18), where $e(m)$ is the roughness of the pipe.

$$\lambda = 0.1 \left(\frac{e}{d} \right)^{0.23} \quad (18)$$

Since the flow of the main line passes through the parallel heat exchanger, its flow distribution can be expressed by Eq. (19), where $V_{\text{percent}}(n, i)$ (dimensionless) is the proportion of the flow rate of the heat exchanger to the total flow rate of all heat exchangers.

$$V \left(n, i \right) = V(n) \cdot V_{\text{percent}} \left(n, i \right) \quad (19)$$

Through the valve of the parallel branches to adjust the flow distribution between the branches, the resistance loss between the parallel lines are the same, that is, the pressure drop of the branches are the same. The valve of the branch are used to adjust the flow of this branch, and the branch pressure drop can be divided into two parts. The first part is the pressure drop of the heat exchanger, the second is the pressure drop of the valve of the branch. Assuming that the pipe length of the branch where the valve is located is exceedingly short, the pressure drop of the branch can be ignored, and only consider the pressure drop caused by the equivalent length of the valve.

The flow distribution of branch i_1 and branch i_2 is derived by deriving the same pressure drop, and the pressure drop of branch i_1 and branch i_2 are same, expressed by Eq.(20).

$$\Delta P_{ex}(n, i_1) + \Delta P_{branch}(n, i_1) = \Delta P_{ex}(n, i_2) + \Delta P_{branch}(n, i_2) \quad (20)$$

According to the resistance coefficient of valve in the main pipe and the branch of the heat exchanger, the flow distribution can be solved, so the proportion of the flow rate of the branch of the heat exchanger i in sub-period n is shown is Eq.(21).

$$V_{\text{percent}}(n, i) = \frac{V(n, i)}{V(n)} = \frac{1}{\sum_{i=1}^{n_{\text{exchanger}}} \frac{1}{\sqrt{\frac{8}{\pi^2 d^4} \xi \left(n, i \right) + \frac{8N^2(i)}{m^2 \pi^2 d_{ex}^4} \left(\lambda \frac{l_{ex}}{d_{ex}} N(i) + \xi N(i) + 1.5 \right)}}}} \quad (21)$$

The intersection point can be found immediately by combining the characteristic curve Eq.(4) of the pipeline and the characteristic curve Eq.(10) of the pump, and the intersection of the two curves is the working point of the pump.

2.2.2. Cooler model

Each heat exchanger undertakes a fixed heat transfer task, using heat to constrain the flow rate of circulating water, and

Eq.(22) is used to describe the heat load of the heat exchanger to meet the heat exchange requirements.

$$Q(n, i) > Q_{required}(i) \quad (22)$$

The heat load required for the cooling of the hot stream can be calculated by Eq.(23), where $C_{ph}(J/(kg \cdot K))$ is a constant, for the constant pressure heat capacity of hot stream, $M_h(kg/s)$ refers to the mass flow of hot stream, and T_1 is the inlet temperature of hot stream, and T_2 is the temperature of hot stream after cooling.

$$Q_{required}(i) = c_{ph} M_h (T_1 - T_2) \quad (23)$$

The heat load $Q(n, i)$ of circulating water can be calculated by the NTU method, which has three important parameters, the ratio of the heat capacity flow rate of the hot and cold fluid R , NTU and ε . where $R(n, i)$ is the ratio of the smaller fluid heat capacity flow rate to the larger heat capacity flow rate of the heat exchanger i in sub-period n , and $R(n, i) < 1$, which can be calculated by Eq.(24).

$$R(n, i) = \frac{(M \cdot c_p)_{small}}{(M \cdot c_p)_{large}} \quad (24)$$

NTU is the number of heat transfer units, which can be calculated by Eq.(25), where K is the convective heat transfer coefficient outside the heat exchanger tube, which can be calculated by Eq.(26), and the convective heat transfer coefficient in the tube is related to the flow rate of the fluid in the tube, which can be calculated with Eq.(27). $A(m^2)$ is the heat exchange area of the heat exchanger, which is determined by the heat exchanger itself. Circulating water takes the tube side, and the hot stream takes the shell side.

$$NTU = \frac{KA}{(M \cdot c_p)_{small}} \quad (25)$$

$$\frac{1}{K} = \frac{1}{\alpha_o} + \frac{1}{\alpha_i'} \quad (26)$$

$$\alpha_i'(n, i) = \alpha_i(n, i) \left[\frac{u(n, i)}{u_o} \right]^{0.8} \quad (27)$$

Eq.(28) ε is the heat transfer efficiency, which can also be expressed by Eq.(29), and the ε is the ratio of the heat load Q to the theoretical maximum heat transfer Q_{max} , indicating the heat transfer efficiency of the heat exchanger.

$$\varepsilon = \frac{1 - e^{(NTU \cdot (1-R))}}{R - e^{(NTU \cdot (1-R))}} \quad (28)$$

$$\varepsilon = \frac{Q}{Q_{max}} \quad (29)$$

The theoretical maximum heat transfer $Q_{max}(w)$ can be calculated by Eq.(30).

$$Q_{max} = (M \cdot c_p)_{small} (T_1 - t_1) \quad (30)$$

Combining Eq. (29) (30) yields the heat load Q of the heat exchanger i in sub-period n .

$$Q(n, i) = \varepsilon \cdot (M \cdot c_p)_{small} (T_1 - t_1) \quad (31)$$

3. Case study

In the Case study section, this paper examines an adapted real case. In this case, there are 21 hot streams that need to be cooled. Exchanging heat through the heat exchanger and circulating water. The summer water temperature of 30 °C requires a circulating water volume of about 3 m³/s. There are 5 20SH-19A pumps including the backup pumps running in parallel. The circulating water is divided into 21 branches and enters the heat exchangers to exchange heat with the hot stream. The temperature at which circulating water enters the heat exchanger varies with the seasons, with lower temperatures in winter and higher temperatures in summer. The heat transfer model of the cooling tower is not considered, and the inlet temperature of the circulating water is set to reflect the 12 month ambient temperature difference. The operation of the project is simulated in 12 months of a year, with different inlet water temperatures for each month. In the case of valve control, the pumps are CSPs, which discontinuously increase the flow that the pump system can provide by parallel pumping. The characteristic curve of the pump is adjusted by the parallel pump, and the characteristic curve of the pipeline is adjusted by adjusting the opening of the 22 valves to obtain the working point of the pump. Under VFD control, some pumps are installed with VFDs and become VSPs. The flow is continuously regulated by CSPs, VSPs and 22 valves.

On the basis of the analysis of the case results, the sensitivity analysis of the model is performed. Different ambient temperature differences, different pump types and their combinations, as well as different electricity charges are analyzed. In the one-year operation cycle, this paper sets up 4 sets of temperature differences to simulate CCWSs in different geographical locations, and sets up 4 pump system configurations (3 types of pumps and 1 combination of them) to simulate the frequency conversion effect of different pump systems, and sets 4 kinds of electricity prices to analyze the VFD energy saving effect of pumps under different electricity prices.

In this MINLP model, this paper is optimized by genetic algorithm, and the crossover probability and variation probability in GA are 0.9 and 0.2, respectively, and the population number is 100. GA iterates 1000 times. The optimization is implemented by Python 3.9.7. The processor is AMD Ryzen 7 5800 H with Radeon Graphics 3.20 GHz. Due to the large range of values of variables, theoretically the resistance coefficients of valve in the main pipe and the branch can reach infinity, resulting in too long running time of the program, so according to the results, the value of the independent variables is preliminarily optimized, and it is found that the optimal solution of the resistance coefficient of valve in the main pipe is in the range of 0–2500, and the resistance range in the branch is within 0–5, so the range of resistance coefficients is narrowed. After preliminary optimization, the value range of the independent variable is determined as follows: the resistance coefficient of valve in the main pipe is 0–2500, the resistance coefficient of valve in the pipeline where the heat exchanger is located is 0–5, the speed ratio k of the speed regulating pump is 0.7–1.3, in addition to two 0–1 variables, the first is the opening and closing of the pump, if the value is 0, this means that the pump is closed. If the value is 1, it means that the pump is in operation. The second is whether the pump is equipped with a VFD.

Table 1 – Valve control related data.

Month	Total Flow rate (m ³ /s)	Number of CSPs	Flow rate (m ³ /s) of each pump	CSPs efficiency	Pressure drop (kPa)
1	1.00	2	0.50	84.9 %	314
2	1.17	2	0.58	82.5 %	284
3	1.34	2	0.67	74.8 %	251
4	1.49	3	0.50	84.9 %	315
5	1.65	3	0.55	84.1 %	296
6	2.07	3	0.69	72.0 %	242
7	2.44	4	0.61	80.8 %	275
8	2.00	3	0.67	74.9 %	252
9	1.54	3	0.51	84.8 %	309
10	1.39	2	0.69	71.2 %	240
11	1.33	2	0.67	74.9 %	252
12	1.17	2	0.58	82.5 %	284

3.1. Result and analysis of the case

In the case, the temperature difference is ranging from 18.0 to 30.0 °C, and 20SH-13A pump is chosen. The electricity price is 0.045 \$/kWh. In this case, the economic performance, pump operating status and pipeline status of Valve control and VFD control are compared and analyzed.

3.1.1. Comparison of economic performance

For the total cost comparison of Valve control and VFD control in temperature 18.0–30.0°C with electricity price of 0.045 \$/kWh during one-year operation. As expected, when the pumps are equipped with VFDs, the total cost is significantly less than valve control. The total cost under valve control is 2.53×10^6 \$. The cost of VFD control is 1.91×10^6 \$ and it consists two parts, the operating cost of the pump and the capital cost of the VFDs. They are 1.84×10^6 \$ and 6.25×10^4 \$ respectively. The investment cost of VFDs is exceedingly small, only 3.4 % of the total cost, but it can provide huge economic benefits and reduce the waste of electric energy, and the total saving is 6.2×10^6 \$, about 24.6 %.

3.1.2. Comparison of pump operating status

In this part of the paper, the operating status of the pump is compared and analyzed. Table 1 shows the total cost, flow rate of each CSP, number of CSPs and the efficiency. In general, pump shaft power is positively related to flow, so minimising flow can reduce pump shaft power and therefore overall cost while meeting heat demands. In the case of valve control, the system adjusts the flow through the number of pumps and the valve of the main pipe, reducing the power of the pumps. However, the price of closing the valve to adjust the flow is the reduction of the operating efficiency of the pump and the huge pressure drop, resulting in waste of electric energy. The BEP efficiency of 20SH-13A is 85.0 %, and the corresponding flow rate is 0.52 m³/s. It can also be seen from the table that under the valve control, the pump does not operate at the highest efficiency point, and the efficiency of the pump decreases in some months and deviates from its BEP. At the same time, the pressure drop under valve control is large, ranging from 242 to 315 kPa, and the increase in resistance causes a huge pressure drop.

Fig. 2. illustrates the speed ratio k and the efficiency of 20SH-13A under VFD control. In the left part speed ratio k in Fig. 2., 0 means the pump is off, and 1 means the pump is running and the pump is a CSP, and a decimal between 0.7 and 1.3 means the speed ratio of the VSPs. As can be seen

from the results, a total of 4 pumps have VFDs installed. Among them, in different months, some VSPs reduced the frequency, and some VSPs increased the frequency. In the right part of Fig. 2., The efficiency of VSPs is maintained in a high range. The frequency of the engine is adjusted by the frequency converter, which can continuously adjust the flow rate and ensure that the pump runs near its BEP. At the same time, the total pressure drop of the pipeline every month under VFD control ranges from 177 to 244 kPa, which is nearly 70 kPa lower than valve control.

Fig. 3. is the operating status of pump in valve control and VFD control under the condition of water temperature of 18.0 °C and 30.0 °C. The left side of the figure is 18.0 °C, and the upper part is valve control. At 18.0 °C under valve control, two CSPs are open. And at 18.0 °C under VFD control, two VSPs are open. At 30.0 °C under valve control, four CSPs are open, turning on two more pumps than at 18.0 °C, and at 30.0 °C under VFD control, 4 pumps are turned on, including three CSPs and one VSP, and its k is 1.05. It can be seen that when the water temperature is higher, the number of VSPs is less, and the frequency of the VFD is increased to increase the flow.

3.1.3. Comparison of pipeline status

Fig. 4. is the resistance coefficient of valve in the main pipe and branch under valve control. The left part of Fig. 4. is the resistance coefficient of valve in the branch, and the right part is the resistance coefficient of valve in the main pipe. The resistance coefficient in the some branches are relatively large. Through the analysis of the heat load of the heat exchanger, it is found that the heat exchanger controlled by the valve with a larger resistance coefficient requires a smaller flow to meet its heat demand. While the branch with a smaller resistance needs relatively larger flow rate, and the heat is not easy to meet, accordingly the valve opening is large, which is also consistent with the common sense of actual industrial operation. The resistance coefficient of valve in the main pipeline during the period from January to December first decreased and then increased, which was opposite to the change trend of flow. In January, the maximum resistance coefficient reached 1126, indicating that the flow rate was reduced by reducing the valve opening to reduce the power of the pump, and the valve opening was small. It can also be seen from Table 1 that reducing the flow rate by decreasing the valve opening will reduce the efficiency of the pump and increase the pressure drop in the pipeline, resulting in a large

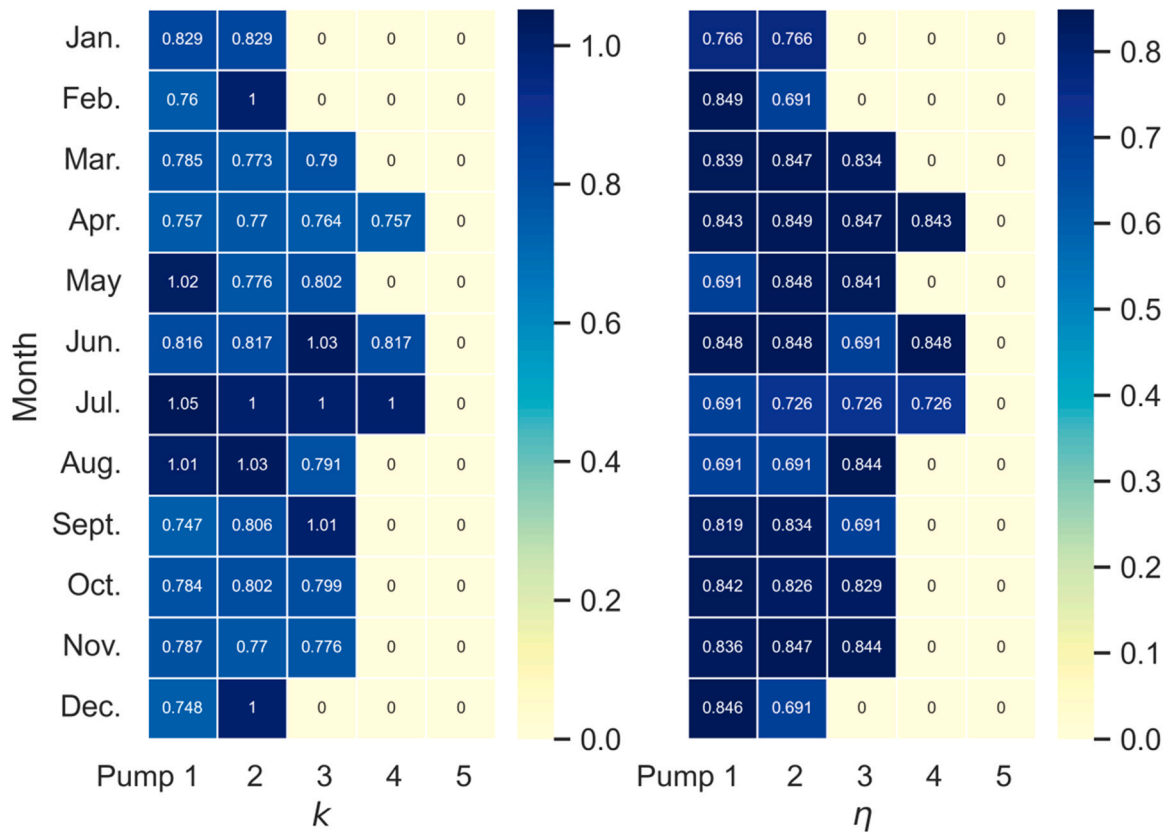


Fig. 2 – VFD control 20SH-13A speed ratio k (left) and efficiency(η right).

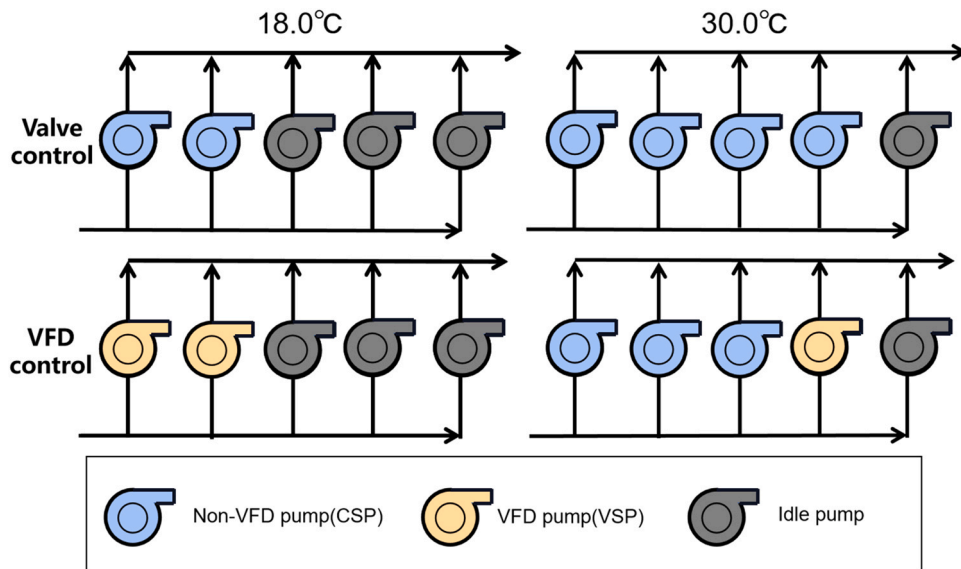


Fig. 3 – 18.0 °C and 30.0 °C valve control and VFD control 20SH-13A pumps operating status.

amount of waste of electrical energy. Reduced pump operating efficiency results in the operating state of CSP deviates from the BEP.

Fig. 5. is the resistance coefficient of valve in the branch (left) and main pipe(right) under VFD control. Compared with valve control, the resistance of valve in the branch under VFD control is also smaller, consequently the operation of the branch valve is less. The values of resistance coefficient of valve in the main pipeline are all around 0. This illustrates that in the case of VFD control, in different months, the

valves of the main pipe are almost always in a fully open state. In actual operation, it is only necessary to ensure that the valve is fully opened, and the flow is adjusted by VFD, which avoids manual valve adjustment and reduces the pressure drop of the pipeline.

3.2. Sensitivity analysis

In this section, sensitivity analysis of the model is carried out on the temperature difference, the pump type and the level

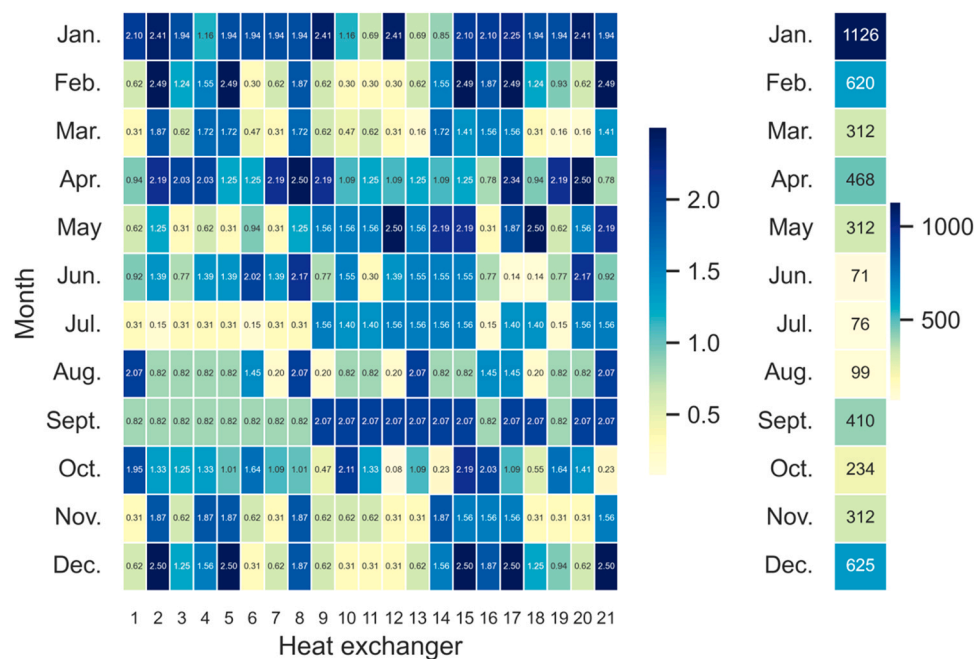


Fig. 4 – Valve control resistance coefficient of valve in the branch(left) and main pipe(right).

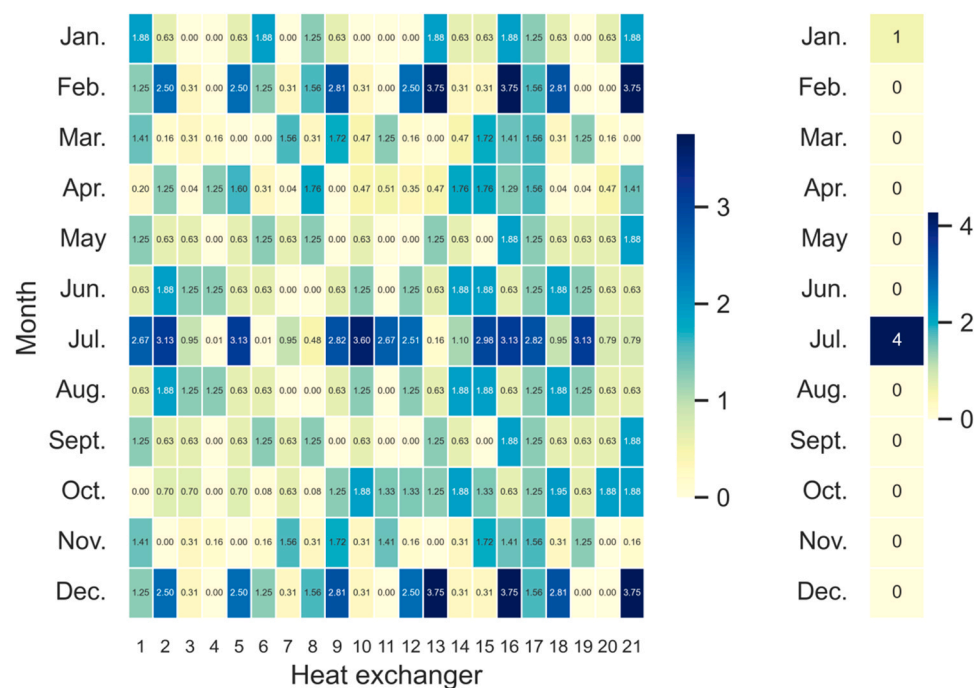
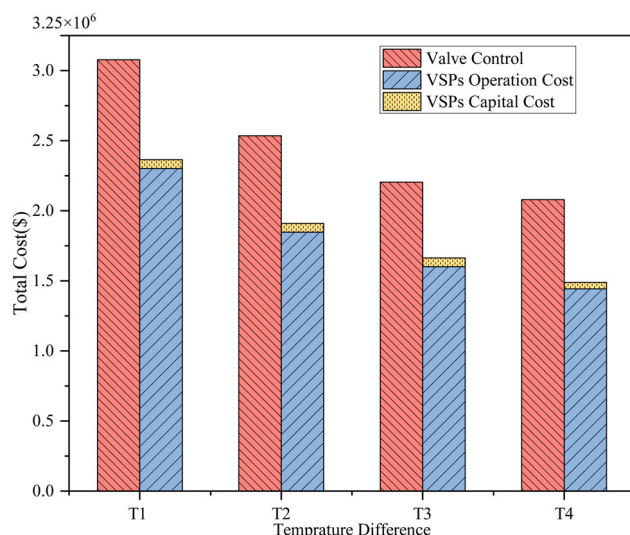


Table 2 – Range of changes in outer conditions in sensitivity analysis.

Temperature difference (°C)	Pump	Electricity price (\$/kWh)
24.0–30.0	12SH-13	0.015
18.0–30.0	20SH-13A	0.030
12.0–30.0	24SH-19	0.045
6.0–30.0	4,3,3 combination	0.090

**Fig. 6 – Cost comparison of valve control and VFD control in 4 temperature differences with 20SH-13A with electricity price of 0.045 \$/kWh.**

was analyzed, and a quantitative basis for whether to use VSPs under various conditions was provided.

3.2.1. Sensitivity analysis of temperature difference

The highest water temperature in July is 30.0 °C, and the water temperature in January is 24.0, 18.0, 12.0, 6.0 °C, respectively.

Fig. 6. is the cost comparison of valve control and VFDs control in 4 temperature differences. The temperature gradually decreases from T1 to T4 in winter, and the lowest is 6 °C, and the total cost is also decreasing. Therefore, in the case of the same temperature in summer, the total cost of the temperature difference from T1 to T4 continues to decrease.

In the four temperature differences, fees and number of installed VFDs are shown in Table 3. VFD control can save 23.2 %, 24.6 %, 24.4 % and 28.2 % respectively, with an average saving of 25.1 %. It can additionally be seen from the results that when the temperature difference increases gradually

from T1 to T4, the energy-saving effect of VFD control is better, and the proportion of cost savings is greater. It can be concluded that in the case of larger temperature difference, the energy saving effect of VFDs is better. Therefore, the carbon emission reduction effect of VSPs is better in areas with large temperature differences.

The fluctuation of temperature difference essentially reflects the variation of circulating water flow demand. As the temperature difference is small, the fluctuation of circulating water flow in different months is accordingly insignificant, the change of CSPs, VSPs and valve opening is relatively tiny, and the repercussion of efficient continuous flow regulation brought by VFD is unobvious. Consequently, in areas such as tropical climates with trivial temperature differences, the flow fluctuation between roses is small. In this instance, the installation of VFD needs to be analyzed more prudently and circumspectly to fully evaluate the necessity and sustainability of VFD. In the area where the temperature difference rose fluctuates greatly, resulting in large flow fluctuations, VFD have the ability to guarantee that the supply head of VSPs is lower, so as to diminish the power consumption of the pump to a greater extent while ensuring the pump efficiency.

3.2.2. Sensitivity analysis of electricity price

In the case of valve control, the pump operating cost is proportional to the electricity price, which means that the effect of different electricity prices on valve control is linear, so it is only necessary to analyze the number of VSPs under VFD control in different electricity prices. Since the flow rate of pump 12SH-13 is smaller, and the rated flow rate is 792 m³/s, the quantity of 12SH-13 is more under the same flow rate. In order to analyze the quantity of VFDs under different electricity prices more clearly, pump 12SH-13 is chosen. Four kinds of electricity prices are set, which are 0.015, 0.030, 0.045, 0.090 \$/kWh respectively. Fig. 7. is the comparison of the total cost of valve control and VFD control under different electricity prices. As the electricity price increases, the total cost of valve control increases linearly, and the energy-saving effect of VFDs is becoming more obvious.

Table 4 displays the number of VFDs installed at different electricity prices. When the electricity price is 0.015 \$/kWh, a total of 3 VFDs are installed in the one-year operation cycle. And when the electricity price is 0.090 \$/kWh, 5 inverters are installed, and there is a difference of 2. It can be seen from the results that when the electricity price is lower, the number of VFDs installed is less, and when the electricity bill is higher, more VSPs are used. When the electricity price increases from 0.015 to 0.090 \$/kWh, the energy-saving effect of variable frequency regulation increases from 2.0 % to 27.1 %. Therefore, it can be concluded that the higher the electricity price, the better the energy-saving effect of the VFD.

Table 3 – The number of VFDs and total cost with electricity prices 0.045 \$/kWh with pump 20SH-13A in four temperature differences under valve control and VFD control.

Temperature	Number of VFDs	VFD control total cost (×10 ⁶ \$)	Valve control total cost (×10 ⁶ \$)	Save (×10 ⁶ \$)	Save Percent
24.0–30.0 °C	4	2.36	3.08	0.71	23.2 %
18.0–30.0 °C	4	1.91	2.53	0.62	24.6 %
12.0–30.0 °C	4	1.67	2.20	0.54	24.4 %
6.0–30.0 °C	3	1.49	2.08	0.59	28.2 %

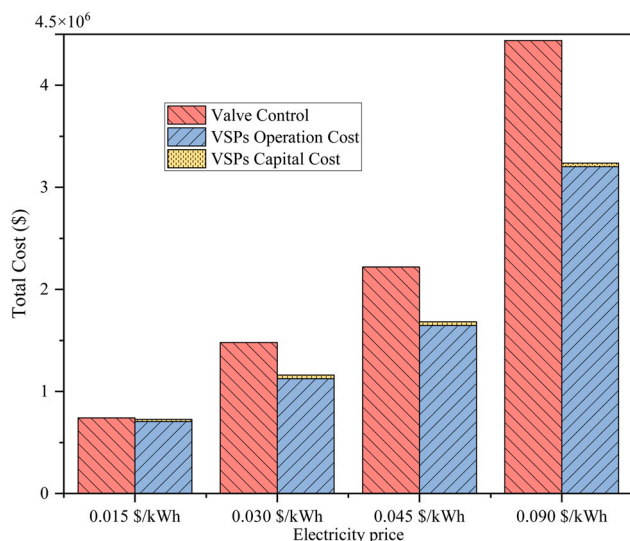


Fig. 7 – Cost comparison of Valve control and VFDs control in 4 electricity prices with 12SH-13 in 18.0–30.0 °C.

In Eq.(1), under the VFD control, the total cost includes two parts, the operation cost of the pump and the investment cost of VFD. The electricity price mainly affects the operation cost of the pump and is irrelevant to the investment cost. From a mathematical analysis, the synergy between total cost and electricity price should be a function of first increasing and then decreasing. Therefore, the concept of balanced electricity price is proposed. There is a balanced electricity price. Before this electricity price, valve control is dominant. The result calculated by the model should be valve regulation without VFD. After balancing the electricity price, VFD control has more advantages. Many parameters in the total cost will affect the balanced electricity price, such as the investment cost of VFD, the frequency of simulated operation of circulating water system, flowrate, etc. In this model, the investment cost of VFD is relatively low, which contributes to the variation of equilibrium electricity price infinitely close to the point where the electricity price is 0. Therefore, in the sensitivity analysis of electricity price, with the increase of electricity price, the energy-saving effect of VFD control is better.

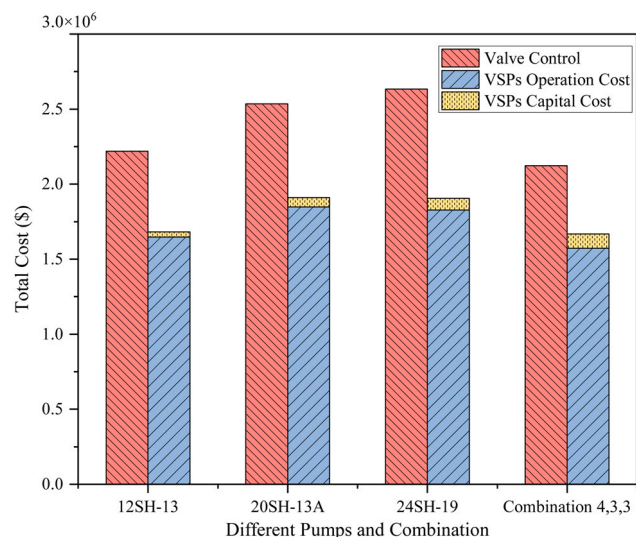


Fig. 8 – Cost comparison of Valve control and VFDs control with different pumps in 18.0–30.0 °C with electricity price of 0.045 \$/kWh.

3.2.3. Sensitivity analysis of pump type and pump combination

Table 5 shows the three pumps data in the sensitivity analysis, and these three pumps and combinations of 4, 3, 3 were selected for sensitivity analysis.

The main difference between the three types of pumps is that the flow rate is different, the supply head difference is intensely small, and the maximum flow rate that can be provided increases in turn. It can be seen from Fig. 8. that under the valve control and VFD control, the total cost of the pump with small flow is smaller, and the combination of the three pumps has the best effect. Compared with valve control, different types of VSPs still have relatively positive energy saving effect.

Table 6 shows the number of VSPs and total cost. The total cost saved by VFDs in the four cases is 25.4 %, 24.6 %, 27.7 % and 21.5 %, respectively. The results show that the model has a superior simulation effect for different types of pumps. Compared with using a single type of pump, the spending of valve control and VFD control is the lowest when using a

Table 4 – The number of VFDs and total cost in four electricity prices in 18.0–30.0°C with pump 12SH-13 under valve control and VFD control.

Electricity price (\$/kWh)	Number of VFDs	VFD control total cost ($\times 10^7$ \$)	Valve control total cost ($\times 10^7$ \$)	Save ($\times 10^7$ \$)	Save Percent
0.015	3	0.72	0.74	0.01	2.0 %
0.030	5	1.16	1.48	0.32	21.6 %
0.045	5	1.68	2.22	0.54	24.2 %
0.090	5	3.24	4.44	1.20	27.1 %

Table 5 – 3 kinds of pump data.

Pump type	12SH-13	20SH-13A	24SH-19
Flow rate(m ³ /h)	792	1872	3168
Supply head(m)	32.2	31	32
Performance curve	$H = 40.90 - 4.88 \times 10^{-3} V + 4.81 \times 10^{-6} V^2 - 1.57 \times 10^{-9} V^3$	$H = 40.09 - 1.8 \times 10^{-3} V + 4.01 \times 10^{-6} V^2 - 1.076 \times 10^{-9} V^3$	$H = 46.73 - 2.5 \times 10^{-3} V + 6.32 \times 10^{-6} V^2 - 4.12 \times 10^{-9} V^3$
Efficiency curve	$\eta = 0.01 + 2.11 \times 10^{-1} V - 1.25 \times 10^{-4} V^2 - 1.15 \times 10^{-8} V^3$	$\eta = 0.00 + 8.92 \times 10^{-2} V - 2.097 \times 10^{-5} V^2 - 1.326 \times 10^{-9} V^3$	$\eta = 0.01 + 5.03 \times 10^{-2} V + 4.726 \times 10^{-6} V^2 - 7.4 \times 10^{-9} V^3$

Table 6 – The number of VFDs and total cost in three pump types and one combination of 4,3,3wo in 18.0–30.0°C with electricity price of 0.045 \$/kWh under valve control and VFD control.

Pump type	Number of VFDs	VFD control total cost ($\times 10^6$ \$)	Valve control total cost ($\times 10^6$ \$)	Save ($\times 10^6$ \$)	Save Percent
12SH-13	5	1.68	2.22	0.56	25.4 %
20SH-13A	4	1.91	2.53	0.62	24.6 %
24SH-19	3	1.90	2.63	0.73	27.7 %
Combination 4,4,3	4	1.67	2.12	0.46	21.5 %

combination of different models of pumps. It can be concluded that larger pumps with VFDs have better energy saving effect and combination pumps yield lower costs.

4. Conclusion

This work presents a new genetic algorithm MINLP optimization model for industrial circulating cooling water system considering VFD. The interaction between the pump and cooler networks was examined. A case study was employed to verify the effectiveness of the model, and the sensitivity analysis of the model is carried out through different temperature differences, pump combinations and electricity prices. The model can determine precise operating parameters for pump system and valve, including the resistance coefficients of valve in the main pipe and branch and the speed ratio k of VSPs, so as to optimize the allocation of flow resources and achieve the effect of energy saving. It can additionally provide guidance on whether to install VFDs for pumps under different environmental conditions, and provide a certain guiding role for the energy saving of CCWSs in practical projects.

The following conclusions can be drawn: (1) Compared with the traditional CSPs valve control, the industrial circulating water MINLP optimization considering VFD proposed in this paper can be used for optimizing CCWSs with different temperature difference conditions, pump combinations, and electricity price levels. (2) VSPs have better energy saving effect when the working condition of the pump deviates greatly from its BEP. (3) For higher electricity prices and greater temperature differences, a pumping system with more VFD-operated pumps save more energy and larger pumps with VFDs have better energy saving effect and pumps combination yield lower costs.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yufei Wang reports financial support was provided by National Natural Science Foundation of China.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.cherd.2022.08.010](https://doi.org/10.1016/j.cherd.2022.08.010).

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